

Developer Ecosystem Analysis 2024

Source: 2024 Stack Overflow Developer Survey



About This Project

Project Scope and Credibility

A data-driven analysis of software development trends built on three real data sources: a jobs API, web scraping, and the 2024 Stack Overflow Developer Survey.

Audience and Purpose

Targets hiring managers, technical leaders, and workforce planners seeking structured, data-backed technology insights.

Credibility Through Methodology

Expanded beyond the base IBM Data Analytics Certificate capstone by creating a full analysis notebook consolidating all data collection methods, cleaning, and processing into one reproducible workflow.

Executive Summary: What the Data Tells Us

JavaScript Usage Dominance

JavaScript leads in current usage with nearly 15,000 active developers, but growth appears to be stabilizing.

TypeScript Rising Popularity

TypeScript has rapidly risen to third place in desired languages, reflecting demand for type safety and scalability.

AWS Cloud Leadership

Amazon Web Services leads cloud infrastructure usage with over 10,800 users, dominating enterprise and startup markets.

PostgreSQL Adoption Pressure

PostgreSQL is the top desired database, showing future migration trends exceeding current usage levels.



Business Question & Dataset

Which technologies are developers using today and which do they want to use next year?



Defining the Business Question

Focuses on current and future technology usage to reveal industry trends and inform strategic planning.



Dataset Overview

Uses 2024 Stack Overflow Developer Survey data covering programming languages, databases, cloud platforms, and web frameworks.



Methodology and Validity

Comparing top technologies in current and desired usage to identify actionable insights from real adoption data.

Key Insights: Number of Jobs by State

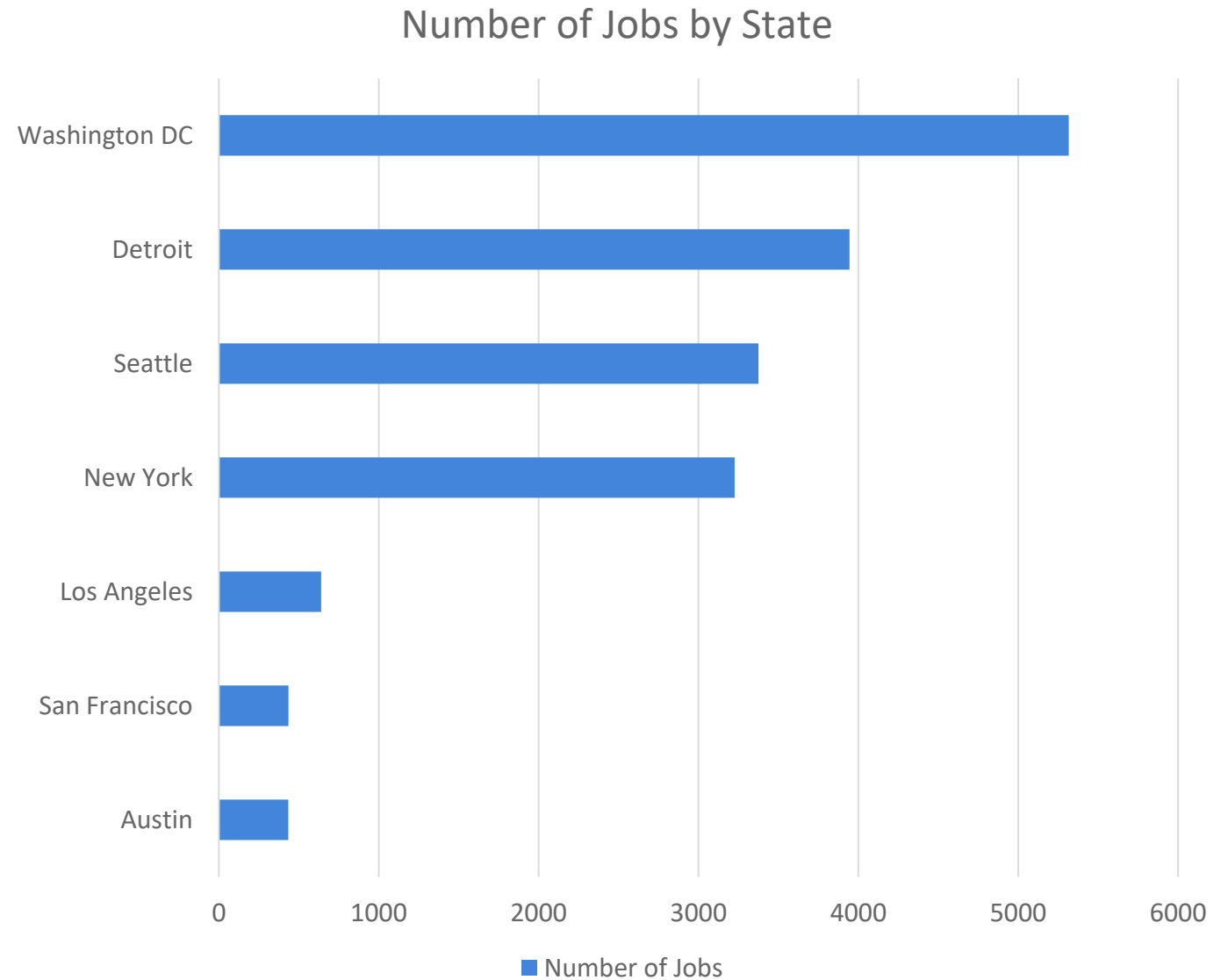
Data collected via IBM Jobs API

Washington DC leads with 5,316 job postings

Detroit ranks 2nd at 3,945 It's a notable tech market often overlooked

Seattle (3,375) and New York (3,226) cluster closely together

West Coast cities lag: LA (640), San Francisco (435), Austin (434)



Key Insights: Average Annual Salary

Data collected via web scraping

Swift leads at \$130,801, driven by iOS/Apple demand

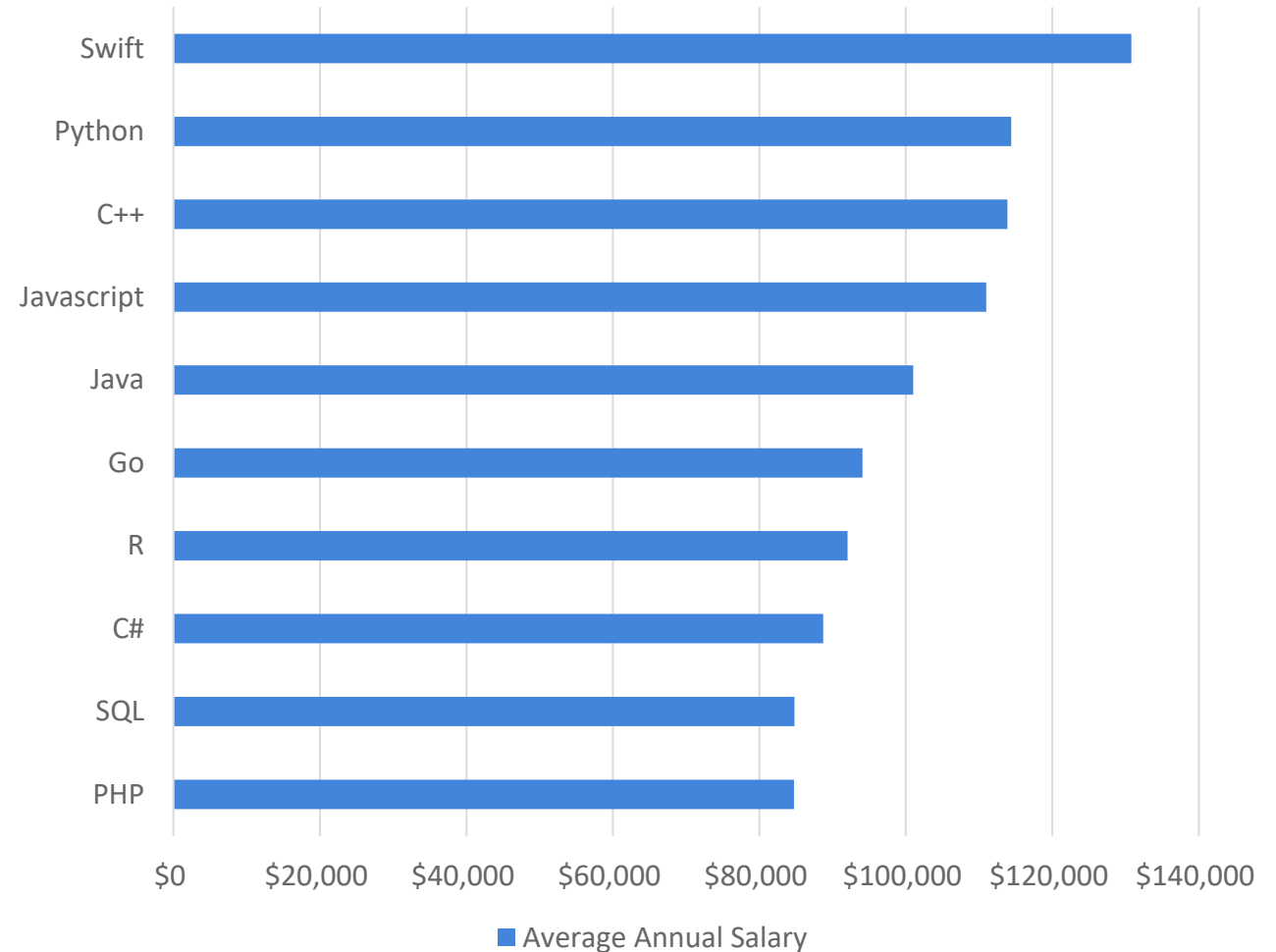
Python (\$114,383) and C++ (\$113,865) follow closely

JavaScript (\$110,981) Most-used language, still pays well

Go (\$94,082) rising demand + solid pay = worth learning

PHP & SQL (\$84-85k) (*SQL demand is near-universal*)

Average Annual Salary by Programming Language



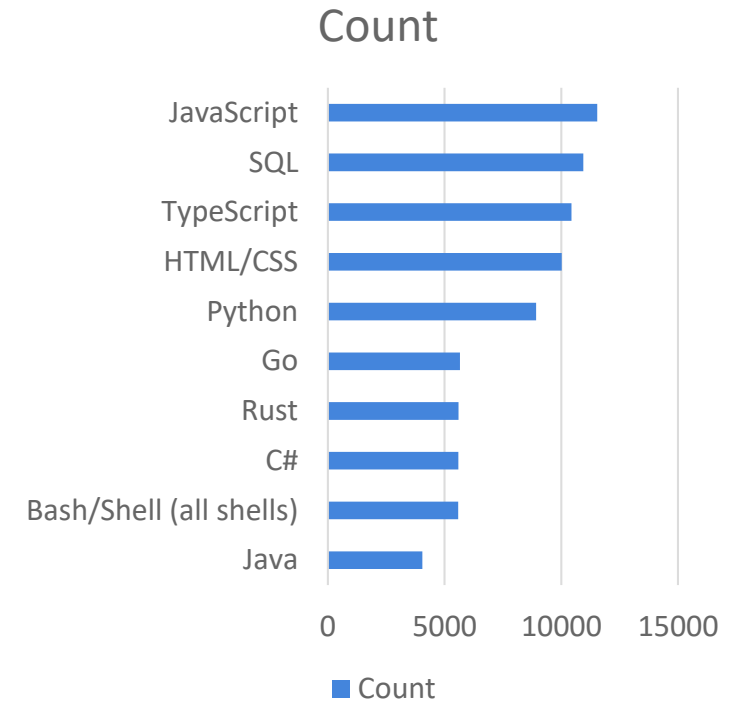
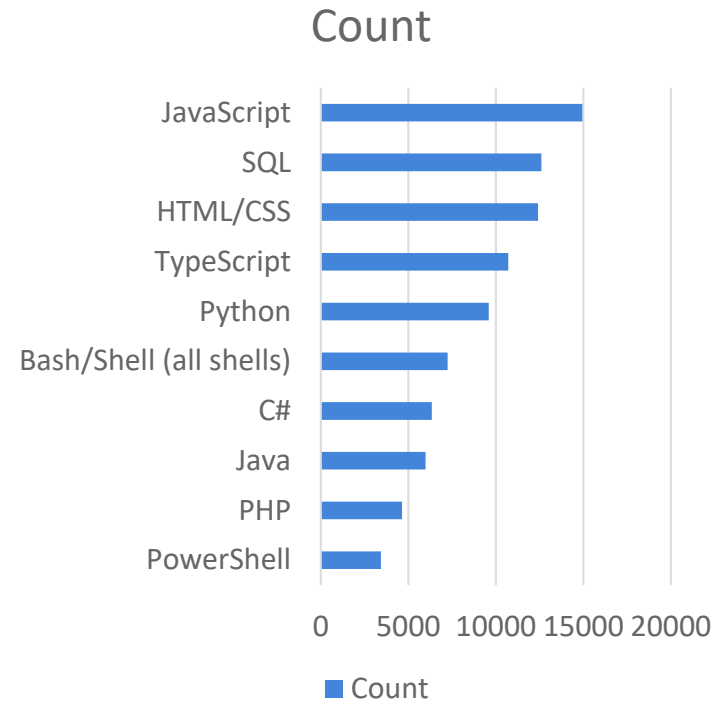
Key Insights: Programming Language Trends

Current (2024):

- JavaScript — 14,943
- SQL — 12,602
- HTML/CSS — 12,410
- TypeScript — 10,709
- Python — 9,590

Desired (2025):

- JavaScript — 11,541
- SQL — 10,944
- TypeScript — 10,437 ↑
- HTML/CSS — 10,016 ↓
- Go — 5,661 (*new entrant*)
- Rust — 5,597 (*new entrant*)



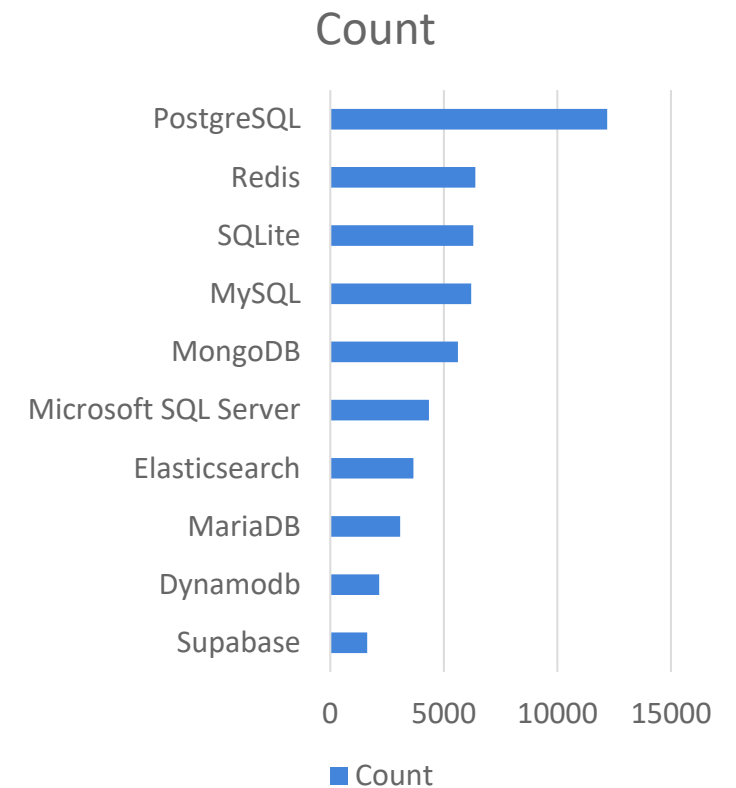
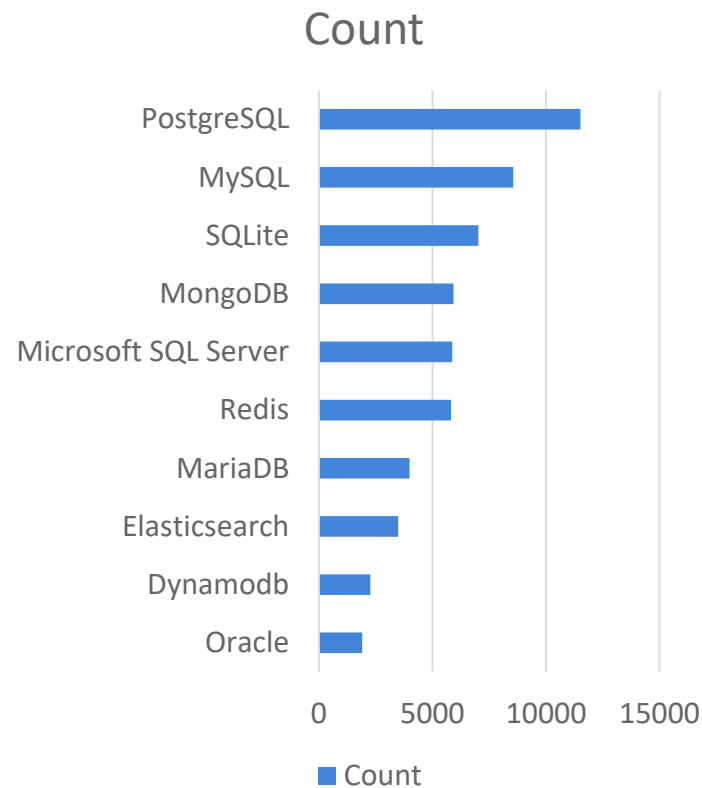
Key Insights: Database Trends

Current (2024):

- PostgreSQL — 11,514
- MySQL — 8,556
- SQLite — 7,021
- MongoDB — 5,930
- Microsoft SQL — 5870

Desired (2025):

- PostgreSQL — 12,193 ↑ (*desired exceeds current*)
- Redis — 6,384 ↑ (*6th → 2nd*)
- SQLite — 6,121 ↓
- MySQL — 5,934 ↓
- Supabase — 1,623 (*new entrant*)



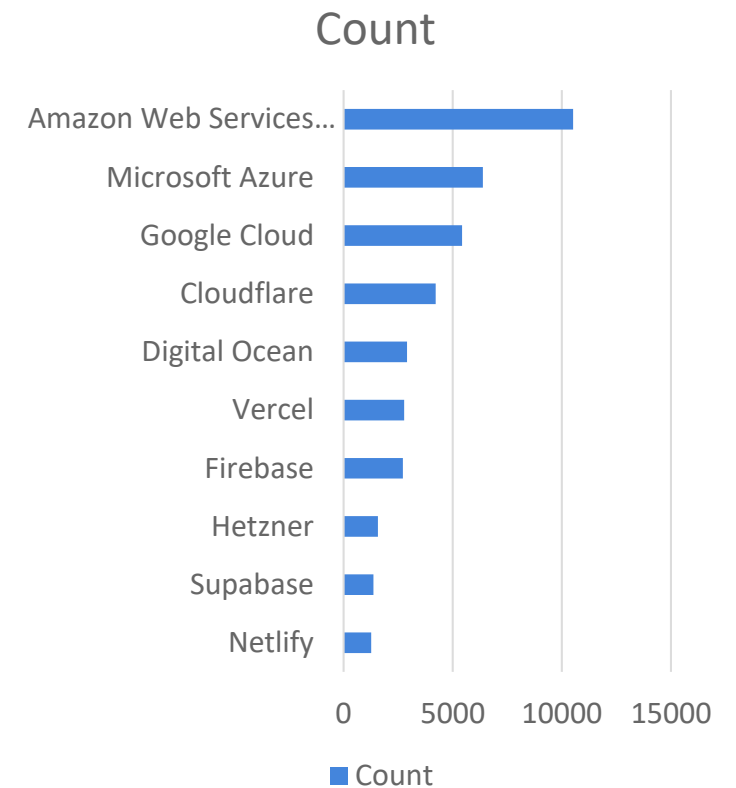
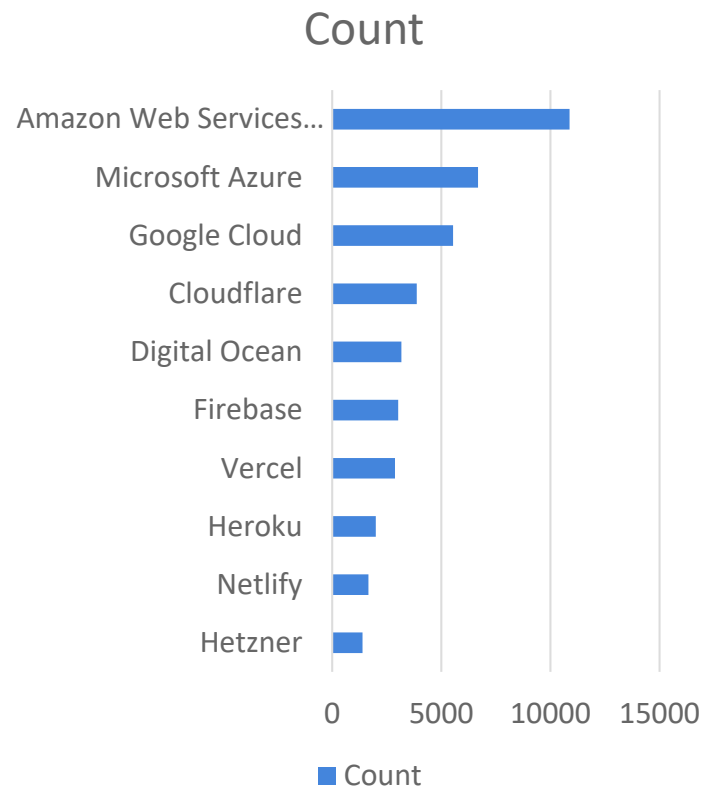
Key Insights: Platform Trends

Current (2024):

- AWS — 10,871
- Microsoft Azure — 6,681
- Google Cloud — 5,537
- Cloudflare — 3,874
- Digital Ocean — 3,170

Desired (2025):

- AWS — 10,516 (*holds #1*)
- Cloudflare — 4,219 ↑ (*biggest % gainer*)
- Supabase — 1,368 (*new entrant*)
- Heroku — dropped out of top 10
- Vercel — 2,776 (*stable*)



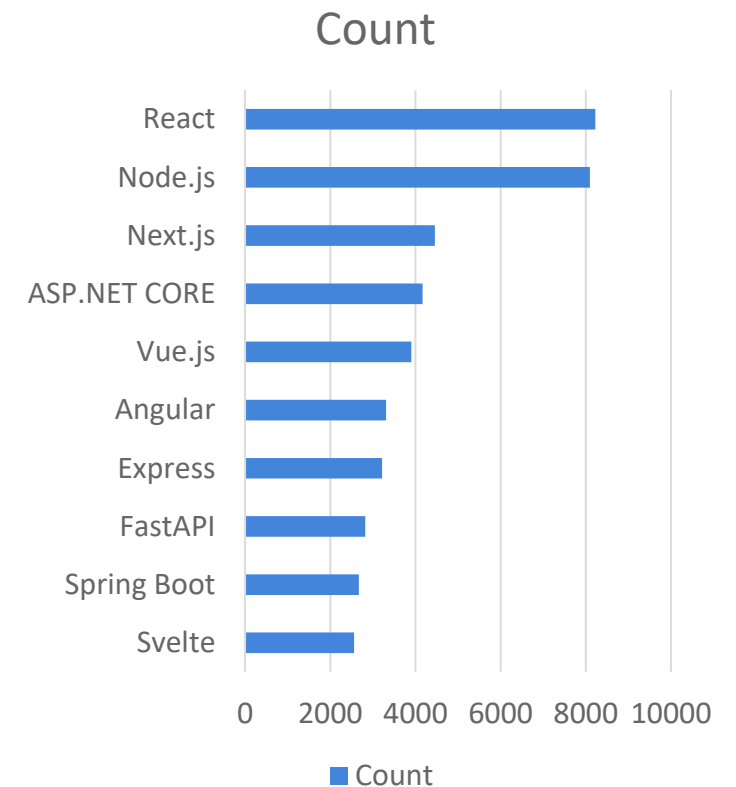
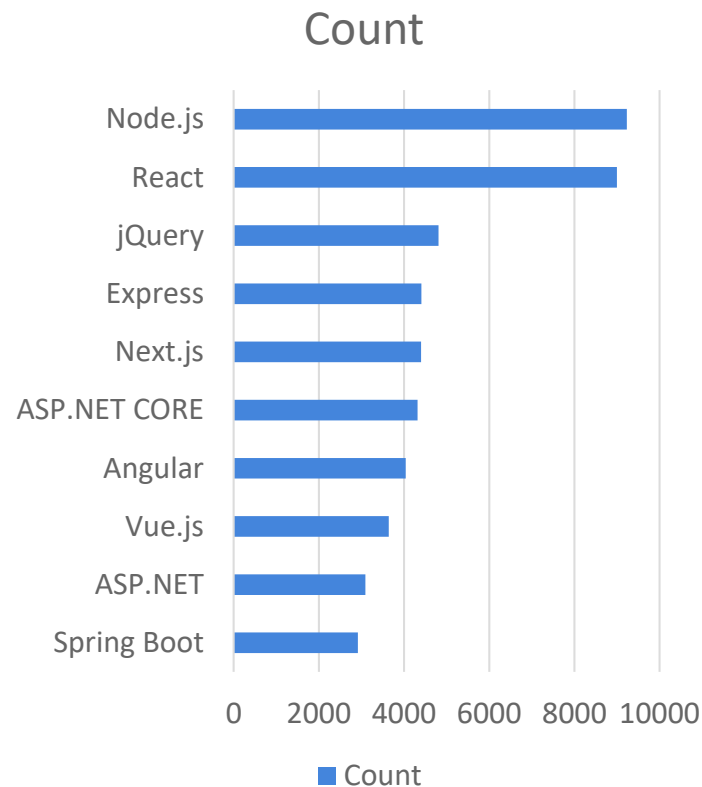
Key Insights: Web Framework Trends

Current (2024):

- Node.js — 9,230
- React — 8,999
- jQuery — 4,810
- Express — 4,408
- Next.js — 4,399

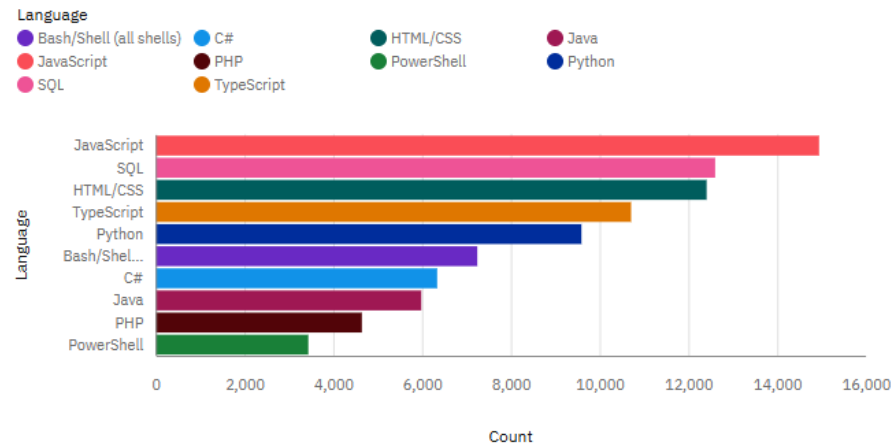
Desired (2025):

- React — 8,223 (*moved to #1*)
- Node.js — 8,096
- Next.js — 4,458 ↑
- FastAPI — 2,823 (*new entrant*)
- Svelte — 2,559 (*new entrant*)
- jQuery — dropped entirely ↓

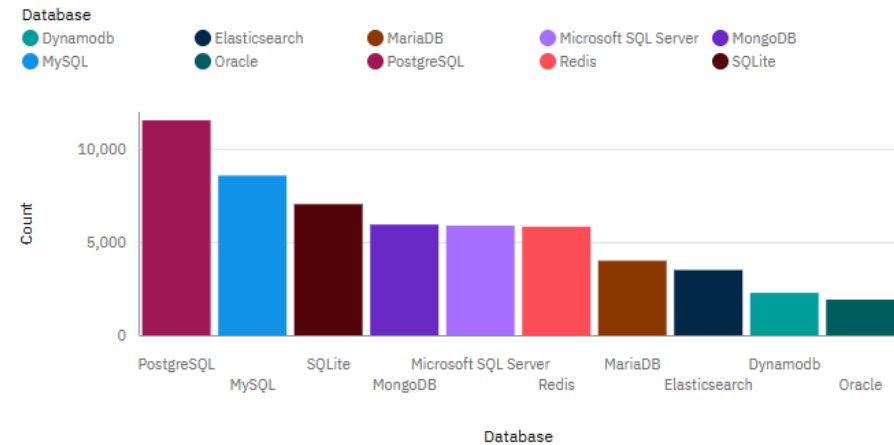


Dashboard Tab 1: Current Technology Usage

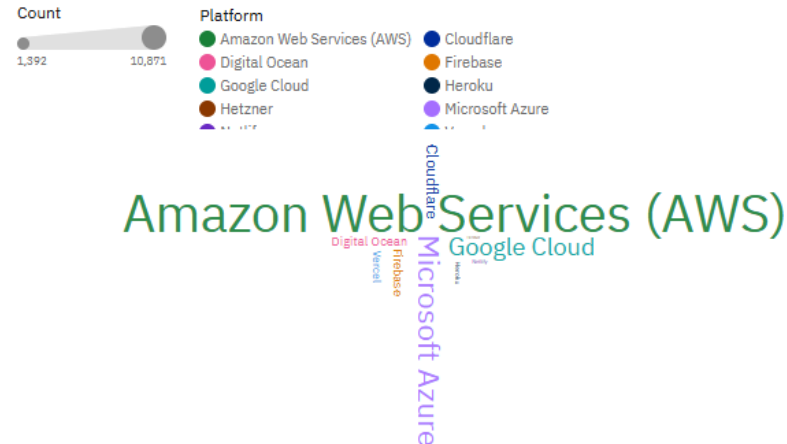
Top 10 LanguageHaveWorkedWith



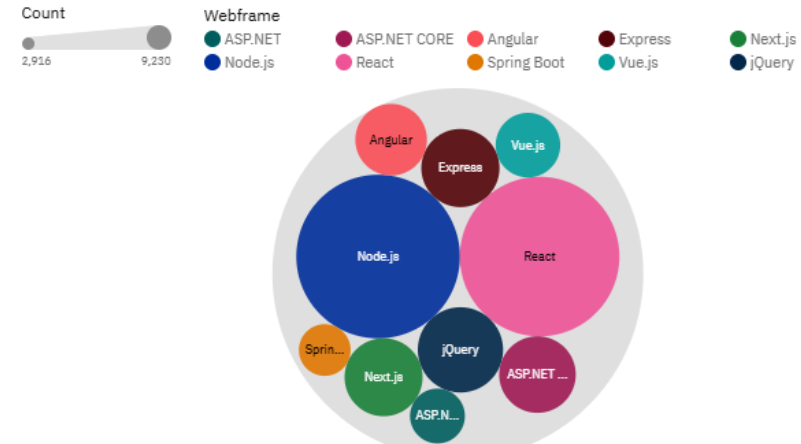
Top 10 DatabaseHaveWorkedWith



Top 10 PlatformHaveWorkedWith

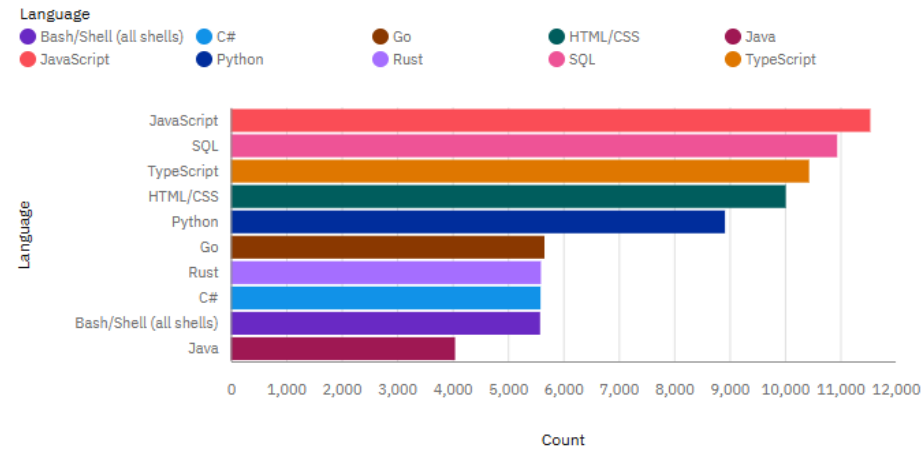


Top 10 WebFrameHaveWorkedWith

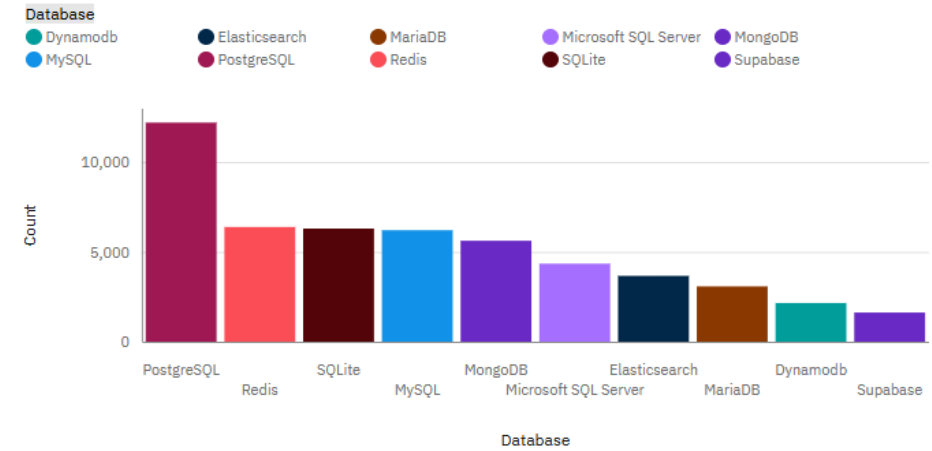


Dashboard Tab 2: Future Technology Trends

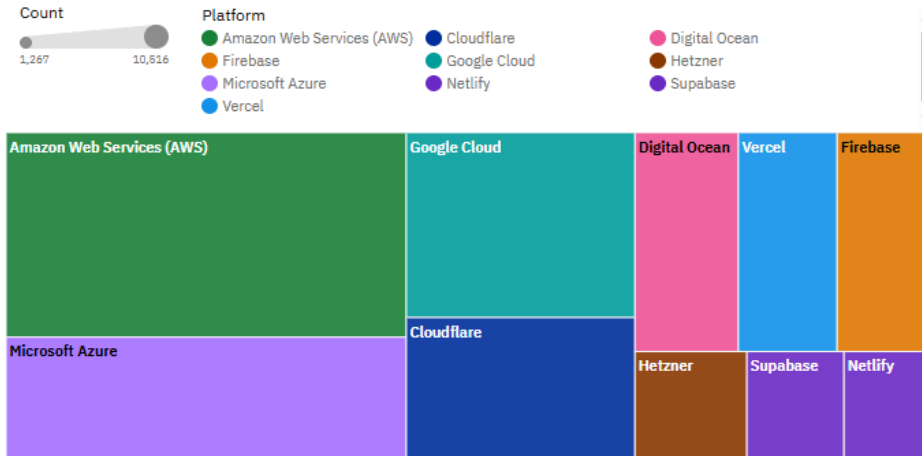
Top 10 Languages Want To Work With



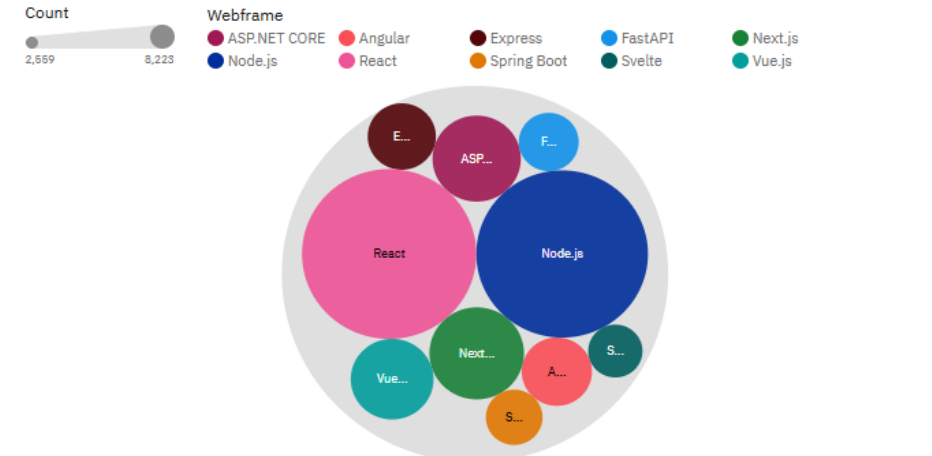
Top 10 Database Want To Work With



Hierarchy for Platform Want To Work With

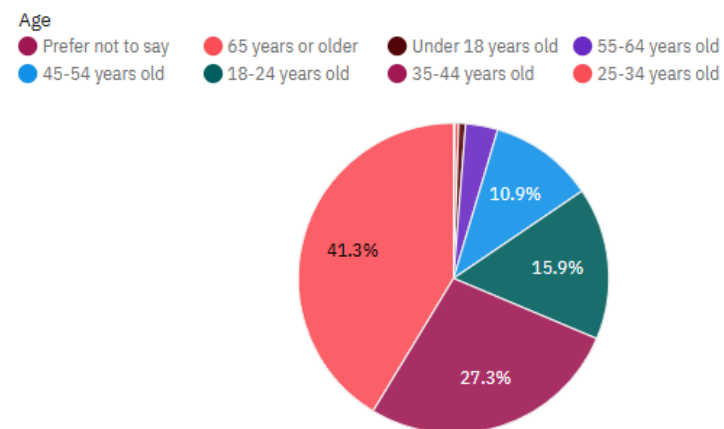


Hierarchy for Webframe Want To Work With

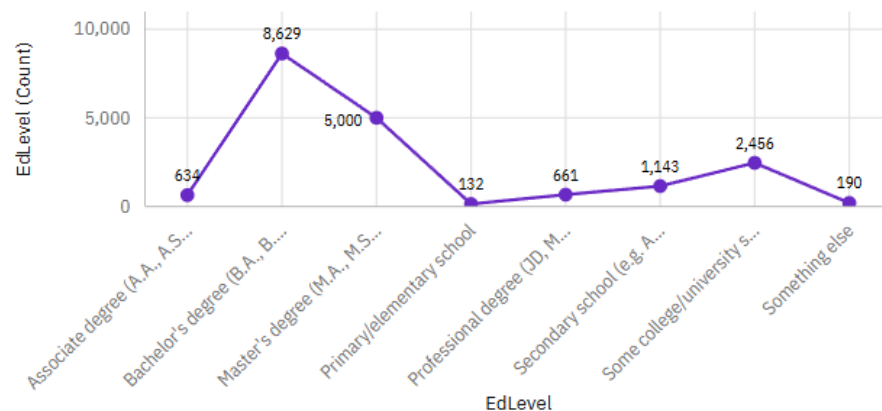


Dashboard Tab 3: Demographics

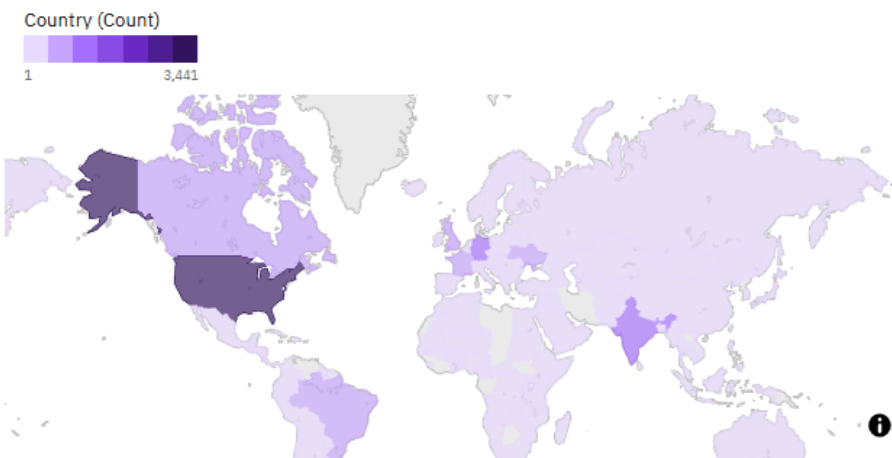
Respondent Distribution by Age



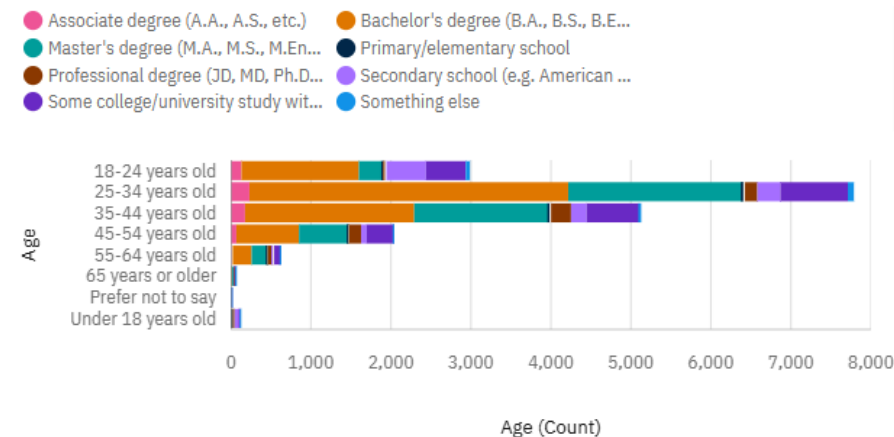
Respondent Distribution by Education Level



Respondent Count by Country



Respondent Count by Age classified by Education Level



Recommendations



Hire for the Core Stack

TypeScript, PostgreSQL, AWS are top-ranked in both current and desired. These are baseline expectations now, not differentiators.



Invest in Emerging Tech

Go, Rust, FastAPI, and Next.js show the highest desired growth. Upskilling here puts teams ahead of the curve.



Reduce Legacy Risk

jQuery, PHP, Oracle, and Heroku are declining in desired usage. Fewer developers want these environments, which compounds into a hiring problem over time.

Skills

API Data Collection (Requests)

Web Scraping (BeautifulSoup)

Data Wrangling & Aggregation
(Pandas)

Data Visualization (Python)

Dashboard Design (IBM Cognos
Analytics)

Business Insight Communication

Conclusion

An End-to-end data analytics project.

From API calls and web scraping to interactive dashboards and business recommendations.

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Linkedin: [Derrick Scott | LinkedIn](#)